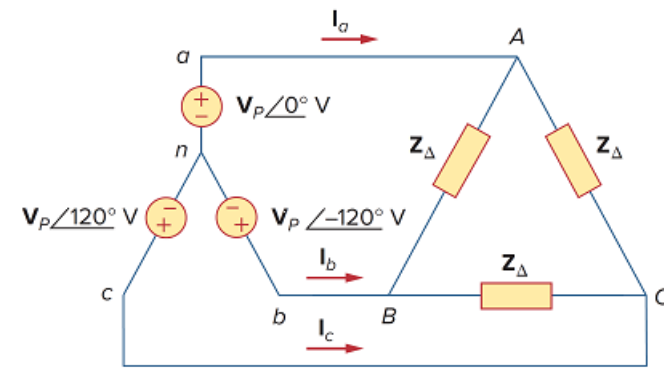


Problem

Using Fig. 12.45, design a problem to help other students better understand wye-delta connected circuits.

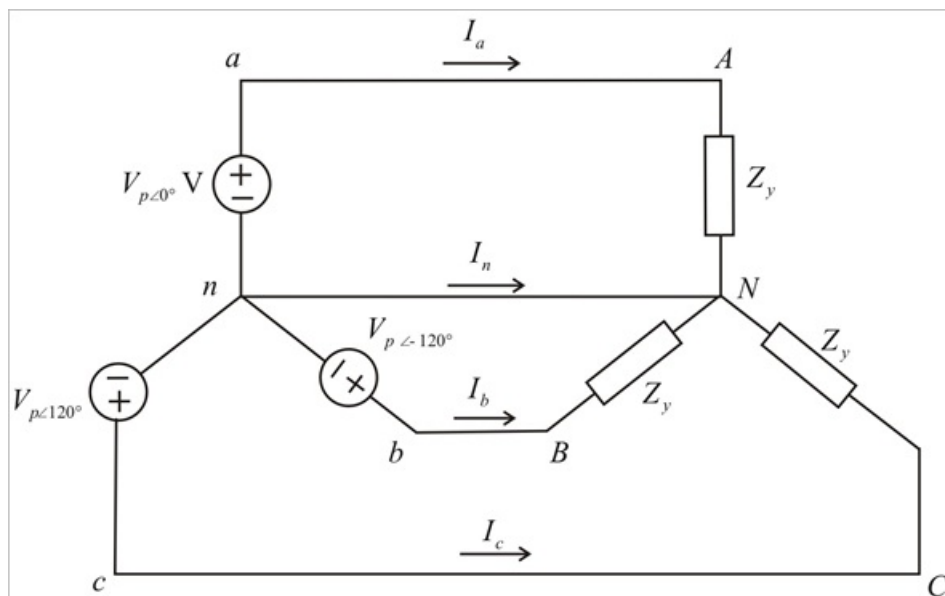
Figure 12.45



Step-by-step solution

Step 1 of 2

We first convert the delta to wye as shown in below circuit



Step 2 of 2

Where,

$$Z_Y = \frac{Z_{\Delta}}{3}$$

From the above circuit,

$$I_a = \frac{V_p \angle 0^\circ}{Z_Y}$$

$$I_b = \frac{V_p \angle -120^\circ}{Z_Y}$$

$$I_c = \frac{V_p \angle 120^\circ}{Z_Y}$$